

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458159.77498 +/- 0.00047 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.113 +/- 0.027
Transit depth (Rp/Rs)^2	1.29 +/- 0.61 [%]
Orbital Inclination (inc)	78.36 +/- 2.78
Airmass coefficient 1 (a1)	0.98 +/- 0.017
Airmass coefficient 2 (a2)	0.0159 +/- 0.005
Scatter in the residuals of the lightcurve fit is	1.72 %
Best Comparison Star	#1 - [435, 46]
Optimal Aperture	4.92
Optimal Annulus	9.82
Transit Duration (day)	0.108 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R█ fit vs published	0.113 ± 0.027 vs 0.1178 (deviation 0.18σ)
Duration fit vs published	0.108 d vs 0.125 d (deviation 1.41σ)
Mid-time O–C	-1573.1 min
Depth SNR	4.2
Residual scatter	1.72 %

Light curve

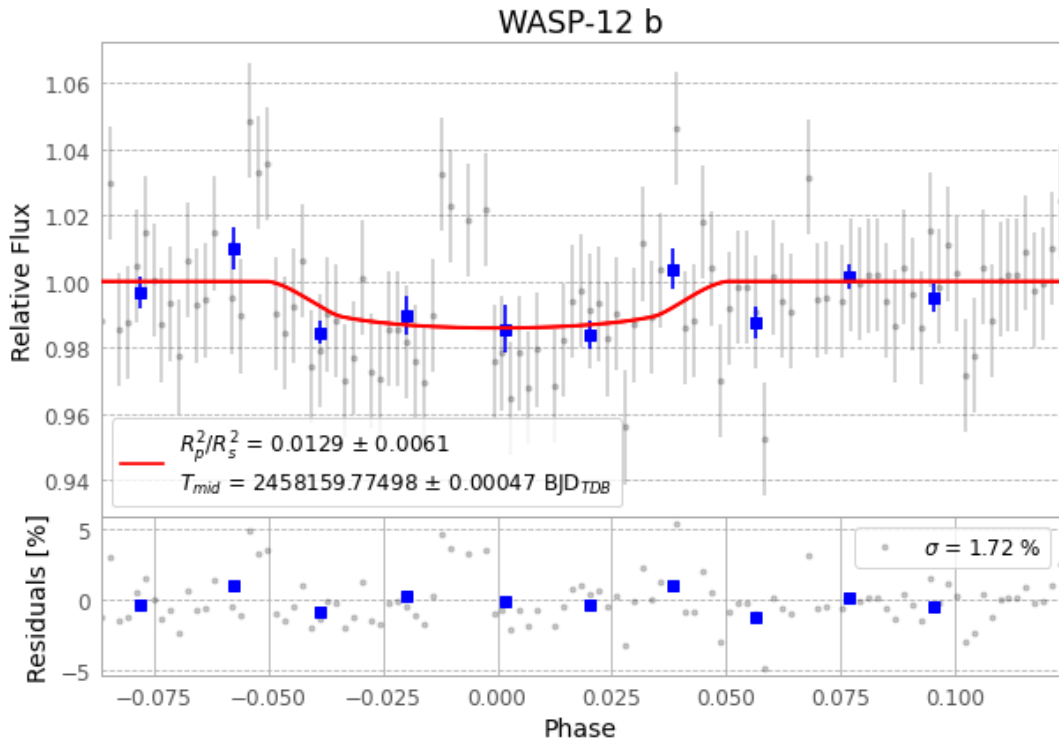


Figure 1 — FinalLightCurve_WASP-12 b_2018-02-09.png (SHA-256 4f76cbeb0c201b9f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 274], 1 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 79371670bf20c90e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

111 FITS frames (73 MB), WASP-12180210041212.FITS → WASP-12180210094212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-180211.log (6f95dc5b1172...), FinalLightCurve_WASP-12 b_2018-02-09.png (4f76cbeb0c20...), FinalLightCurve_WASP-12 b_2018-02-09.csv (3bd34f291f1e...)

Compute resources

Wall time 90.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-22 08:02Z.